Fangzhou Ye

Email: fzy2023@gmail.com / fangzhou.ye@ucf.edu
Personal Website: https://fzy2023.github.io/ Linkedin: www.linkedin.com/in/fangzhou011718/

EDUCATION

Ph.D. in Computer Engineering August 2023 – Present
Department of Electrical and Computer Engineering, University of Central Florida, Orlando, FL, USA.
Thesis advisor: *Dr. Hao Zheng*

Bachelor's in Electronic Science and Engineering September 2019 – June 2023
College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, Zhejiang, China.

RESEARCH INTERESTS

My research interests lie at the intersection of Computer Architecture, Machine Learning, and hardware/software co-design for efficient AI. I focus on improving the computational and memory efficiency of modern AI systems through innovations across algorithms, architectures, and system designs. In particular, my work explores sparsity acceleration, with an emphasis on irregular and unstructured sparsity. Moving forward, I aim to further investigate the intersection of large language models (LLMs), retrieval systems, and sparsity, with a focus on developing efficient architectures and systems for emerging AI workloads.

SKILLS

- Programming languages: C, C++, Python, PyTorch, MATLAB, and Verilog.

MAJOR EXPERIENCES

Teaching/Research Graduate Assistant, University of Central Florida August 2023 – Present
Research Graduate Assistant (Leading Projects)

- Designed a **specific hardware accelerator** with **a novel scheduling policy** for GNN training/inference. Message-passing GNN operations are neither sparse-dense nor general matrix multiplications, so conventional matrix-based optimizations do not transfer. Derived a **Gram matrix formulation** that exposes **data reuse and workload balance simultaneously**, and designed a unified PE architecture around it. Built a **clock-accurate simulator in Python** to validate the design across GNN models. Submitted a paper to DAC 2024 Conference and have been accepted.
- Addressed irregular gradient propagation and the physical constraints of flat on-chip topologies in wafer-scale GNN training. Formulated common network topologies and recast workload partitioning and mapping. Designed a **generative model** in Python for arbitrary topologies rather than relying on topology-specific heuristics. Submitted a paper to the HPCA 2026 Conference and have been accepted.
- Designed a **specific hardware accelerator** to deal with the computation and memory issues in high-order tensor contraction. Proposed a **co-occurrence graph abstraction**, then redefined **tiling, dataflow, and accelerator microarchitecture** on top of it. Submitted a paper to the ISCA 2026 Conference and have been accepted.
- Designed a **specific hardware accelerator** to deal with the **redundancy issues in Higher-order GNNs**. Built a **clique-centric dataflow**, a **K-hop degree partitioning algorithm**, and a compact graph encoding format. Submitted a paper to the MICRO 2026 Conference and have been accepted.
- Worked on irregular sparse issues in scientific computing problems. Reformulated Jacobian computation through the **Schur complement** and designed a **weighted k-core decomposition** with a **Coupled Dependency Graph** to exploit inter- and intra-iteration reuse. Designed the **accelerator architecture** and **dataflow mapping** supporting both computation modes. Submitted a paper to the MICRO 2026 Conference and have been accepted.
- Designed a **CUDA** framework that decouples the cost of multi-hop propagation from the expanded receptive field. Introduced the **K-hop Reachability Matrix**; a KRM-based partitioning algorithm for inter- and

intra-tile reuse; and GPU kernels using **warp-level primitives** to coordinate atomic reductions and cut write contention.

- Built an end-to-end pipeline for redundancy-free graph retrieval and generation. Elevated the unit of reuse from token sequences to **hop-constrained entity–relation paths**, reformulating diverse search, ranking, and reasoning procedures as a shared **join-based bidirectional DFS**. Introduced **Semantic Path KV Reuse**, which materializes recurring semantic paths at stable GPU memory locations with cache-line-aligned layouts and query reordering for L2 locality.

Teaching Graduate Assistant

- **Engineering Analysis and Computation** (EGN 3211), Fall 2024 and Spring 2025. Taught structured programming in C/C++ to first- and second-year engineering students. Led lab sections, held weekly office hours, and wrote and graded programming assignments and quizzes.
- **Signal Analysis and Analog Communication** (EEL 3552), Fall 2024. Ran recitations and office hours covering Fourier and Laplace analysis, sampling, and analog modulation.
- **Digital Design** (EEE 3442), Spring 2025. Supported an introductory course on logic design and the building blocks of digital computers, covering Boolean algebra, combinational and sequential logic, and HDL-based labs. Graded assignments and labs and held office hours.

Student Research Training Program (at national level), Zhejiang University June 2021 – May 2022

- **Highest level** in undergraduate research programs.
- Conducted research on **model segmentation** methods for wireless **federated learning**.
- Designed and implemented **a distributed communication system** using PyTorch in Python.
- Wrote and submitted **a patent** to the China National Intellectual Property Administration.

Internship, Office of Global Engagement, Zhejiang University September 2021 – December 2021

- Handled general affairs as a student assistant.

FULL PUBLICATIONS

Conference Proceedings

[1] **Fangzhou Ye**, Amir Ghazizadeh Ahsaei, Hao Zheng. "Argus: An Efficient Accelerator Design for Sparse Nonlinear Solver", in Proceedings of IEEE/ACM International Symposium on Microarchitecture (MICRO) **(Top-tier Conference)**, Athens, Greece, October 31–November 4, 2026.

[2] **Fangzhou Ye**, Wei Zhang, Ahmed Louri, Hao Zheng. "MOSAIC: An Accelerator Design for Heterophilic Graph Learning", in Proceedings of IEEE/ACM International Symposium on Microarchitecture (MICRO) **(Top-tier Conference)**, Athens, Greece, October 31–November 4, 2026.

[3] Amir Ghazizadeh Ahsaei, **Fangzhou Ye**, Hao Zheng. "Connectivity, Not Coordinates: Accelerating SpMM and High-Order Tensor Contractions via Graph Abstractions", in Proceedings of the 2nd Workshop on Sparsity, the Key Ingredient from HPC to Efficient LLMs (SPICE) at MICRO, Athens, Greece, November 1, 2026.

[4] **Fangzhou Ye**, Shilin Tian, Amir Ghazizadeh Ahsaei, Hao Zheng. "TensorPrism: Rethinking Sparse High-order Tensor Acceleration via Co-occurrence Graph", in Proceedings of the 53rd International Symposium on Computer Architecture (ISCA) **(Top-tier Conference)**, Raleigh, NC, June 27–July 1, 2026.

[5] Shilin Tian, **Fangzhou Ye**, Amir Ghazizadeh Ahsaei, Wei Zhang, Hao Zheng. "PanGaea: A Unified Memory-Efficient Accelerator for Pangenome Chaining and Alignment", in Proceedings of the 63rd Design Automation Conference (DAC) **(Top-tier Conference)**, Long Beach, CA, July 26–29, 2026.

[6] **Fangzhou Ye**, Lingxiang Yin, and Hao Zheng. "Scaling Graph Neural Network Training via Geometric Optimization", in Proceedings of IEEE International Symposium on High-Performance Computer Architecture (HPCA) **(Top-tier Conference)**, Sydney, Australia, January 31–February 4, 2026.

[7] Amir Ghazizadeh Ahsaei*, Lingxiang Yin*, Shilin Tian, **Fangzhou Ye**, Fan Yao, Hao Zheng. "Rethinking Tiling and Dataflow for SpMM Acceleration: A Graph Transformation Framework", in Proceedings of IEEE/ACM International Symposium on Microarchitecture (MICRO) **(Top-tier Conference)**, Seoul, South Korea, October 18–22, 2025. **[*Equal contribution]**

[8] **Fangzhou Ye**, Lingxiang Yin, Amir Ghazizadeh Ahsaei, Hao Zheng. "EGMA: Enhancing Data Reuse and Workload Balancing in Message Passing GNN Acceleration via Gram Matrix Optimization", in Proceedings of the 61st Design Automation Conference (DAC) **(Top-tier Conference)**, San Francisco, CA, June 23–27, 2024.

Patents

[9] Shi Binpu, **Ye Fangzhou**, Liu Shengli, Chen Haoyang, Yu Guanding. "Distributed Communication System and Method," granted by the China National Intellectual Property Administration (CNIPA), 2023. Patent No. CN115150288B.

AWARDS

- Outstanding Graduate Researcher Award, UCF, 2026.
- Future Faculty Laureates (FFLs) Program for Doctoral Students, UCF, 2024–2026.
- MICRO Student Travel Grant, MICRO, 2026.
- ISCA Student Travel Grant, ISCA, 2026.
- Best Reviewer Certificate, TSUSC, 2025.
- MICRO Student Travel Grant, MICRO, 2025.
- ORCGS Doctoral Fellowship, UCF, 2023–2024.
- DAC Young Fellows, DAC, 2024.

PROFESSIONAL SERVICES

Conference Reviewer

- Design Automation Conference (DAC)'24–26
- IEEE International Parallel & Distributed Processing Symposium (IPDPS)'24
- International Conference on Computer-Aided Design (ICCAD)'24–26
- IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)'25–26
- Conference on Machine Learning and Systems (MLSys)'26
- International Conference on Supercomputing (ICS)'26
- International Symposium on High-Performance Computer Architecture (HPCA)'27

Journal Reviewer

- IEEE Transactions on Sustainable Computing (TSUSC)
- IEEE Transactions on Computers (TC)
- Future Generation Computer Systems
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Circuits and Systems I (TCAS-I)
- Computing in Springer Nature

Artifact Evaluation Committee

- MICRO'26 Artifact Evaluation PC Member